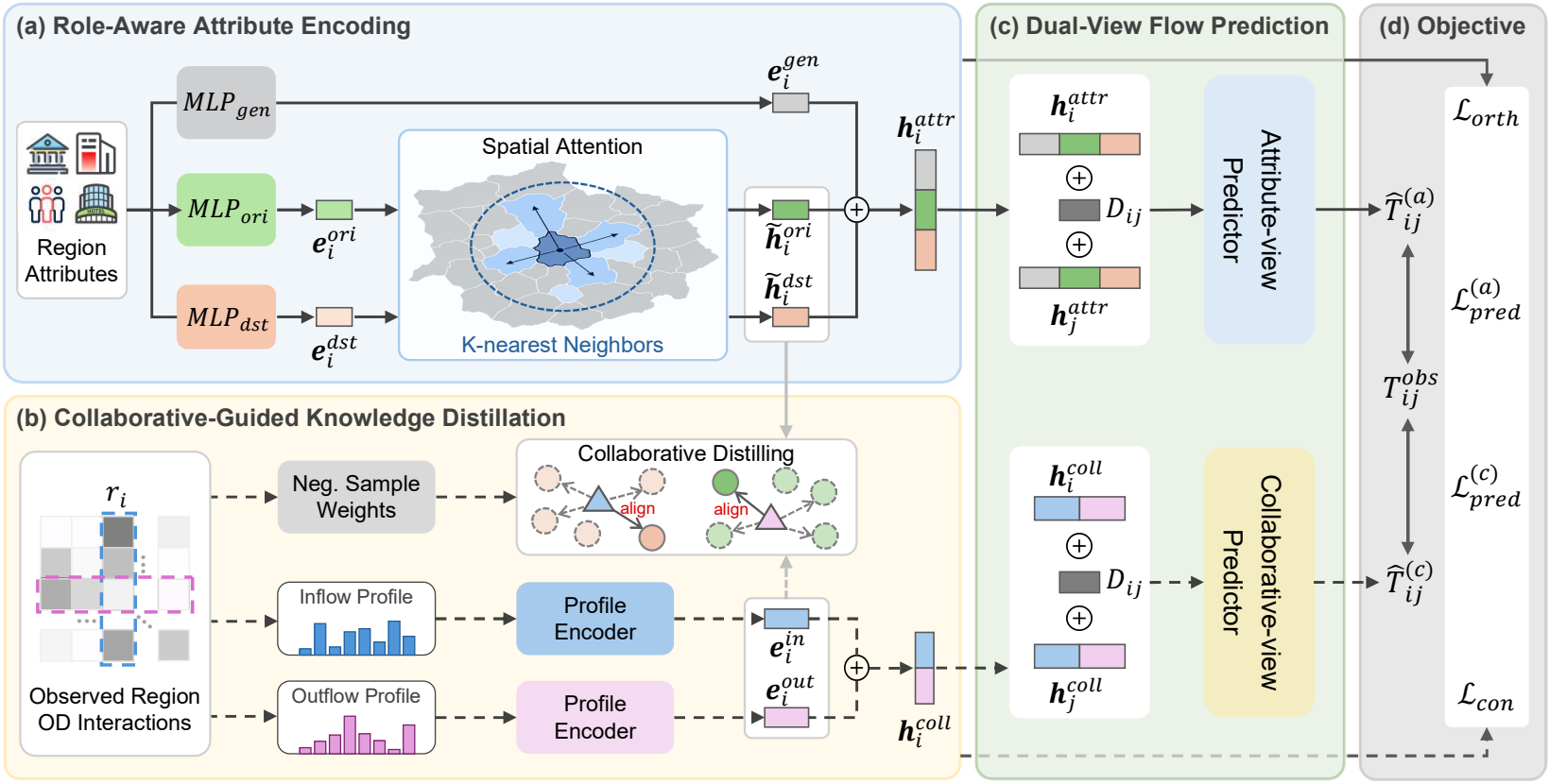

(a) Role-Aware Attribute Encoding
Region Attributes
MLP_{gen}
MLP_{ori}
MLP_{dst}
e_i^{gen}
e_i^{ori}
e_i^{dst}
Spatial Attention
K-nearest Neighbors
$\tilde{h}_i^{ori}$
$\tilde{h}_i^{dst}$
h_i^{attr}
(b) Collaborative-Guided Knowledge Distillation
r_i
Observed Region OD Interactions
Neg. Sample Weights
Inflow Profile
Outflow Profile
Profile Encoder
Profile Encoder
Collaborative Distilling
align
align
e_i^{in}
e_i^{out}
h_i^{coll}
(c) Dual-View Flow Prediction
h_i^{attr}
D_{ij}
h_j^{attr}
Attribute-view Predictor
h_i^{coll}
D_{ij}
h_j^{coll}
Collaborative-view Predictor
(d) Objective
$\mathcal{L}_{orth}$
$\widehat{T}_{ij}^{(a)}$
$\mathcal{L}_{pred}^{(a)}$
T_{ij}^{obs}
$\mathcal{L}_{pred}^{(c)}$
$\widehat{T}_{ij}^{(c)}$
$\mathcal{L}_{con}$